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Origin and evolution of Lord Howe Island, Southwest Pacific Ocean

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ABSTRACT

Lord Howe Island is the eroded remnant of a large shield volcano. Tholeiitic lavas of the North Ridge Basalt comprise the main shield building phase and were erupted about 6.9 Ma ago. The Boat Harbour Breccia probably formed within the throat of the volcano and, together with the North Ridge Basalt, is intruded by numerous basaltic dykes, which grade into a cone sheet complex near the main vent. Large scale collapse of the summit area of the volcano produced a caldera which was filled rapidly by lavas of the Mount Lidgbird Basalt some 6.4 Ma ago, bringing to a close the volcanic history of Lord Howe Island. The shield volcano thus was built during a short interval in the late Miocene.

Palaeomagnetic data show that the North Ridge Basalt and the Mount Lidgbird Basalt were erupted during periods when the geomagnetic field had normal polarity, and that their formation was separated by at least one interval of reversed polarity when the dykes and cone sheets were emplaced. The directions of magnetisation for the lavas and intrusives are such that, palaeomagnetically, no movement of Lord Howe Island is detected since its formation.

Lord Howe Island is the subaerial part of a large seamount which lies at the southern end of a northerly-trending line of volcanic seamounts extending for more than 1000 km. The Lord Howe seamount chain probably was produced by movement of the Australian lithospheric plate over a magma source or hot spot located below the plate within the upper mantle. Other data suggest that the Australian plate is moving N at about 6 cm/a and from this it is predicted that the seamount underlying Nova Bank, at the northern end of the chain, was constructed by volcanic activity about 23 Ma ago. Similarly, if volcanism were to occur now in the Lord Howe seamount chain we predict that its location would be about 400 km S of Lord Howe Island.

INTRODUCTION

Lord Howe Island is located at 31°33'S, 159°05'E, some 600 km E of the Australian continent, and about 800 km ENE of Sydney (Fig. 1). Lord Howe Island and Balls Pyramid, which lies 25 km to the SSE, are volcanic culminations of a large seamount that rises from ocean floor at least 1800 m deep (Fig. 2). This seamount is elliptical in plan, with a major axis (NNW-SSE) of about 65 km and minor axis (ENE-WSW) of 25 km, at the 1000 m isobath (Fig. 2).

Lord Howe Island is built adjacent to the boundary between the two major physiographic features of the region, the Lord Howe Rise and the Tasman Basin (Fig. 1). The Lord Howe Rise is a continental crustal block that became separated from the Australian continent through the development, by seafloor spreading processes, of new oceanic crust that forms the deep Tasman Basin (Hayes & Ringis, 1973; Burns et al., 1973; McDougall & van der Lingen, 1974; Weissel & Hayes, 1977; Weissel et al., 1977). This episode of seafloor spreading occurred during the late Cretaceous and early Tertiary, between about 80 and 60 Ma ago.

The island lies near the southern end of a northerly-trending line of coral-capped guyots, known as the Lord Howe seamount chain, and about 300 km E of another well-defined, northerly striking line of seamounts, the Tasmantid Guyots (Standard, 1961; Conolly, 1969; Vogt & Conolly, 1971; Slater &

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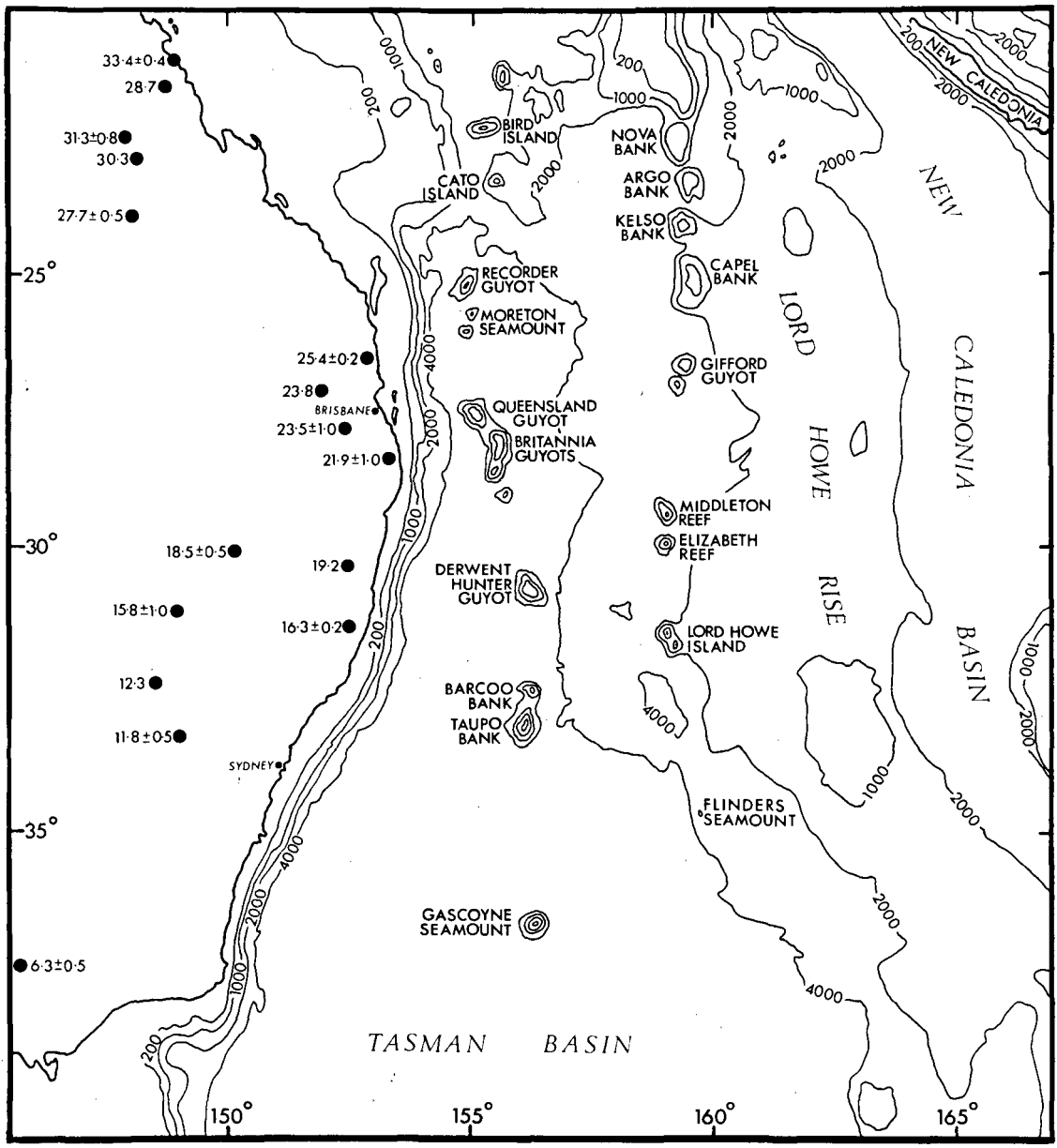


Fig. 1. Bathymetric map of the Tasman Sea, after Mammerrickx et al. (1971), showing location of Lord Howe Island and major topographic features of the region. Isobaths in metres. Filled circles in eastern Australia show locations of central volcanoes and their average K-Ar ages, after Wellman & McDougall (1974), recalculated to the ^{40}K decay constants recommended by Steiger & Jäger (1977).

Goodwin, 1973), which are built up from the floor of the Tasman Basin (Fig. 1).

Lord Howe Island is about 10 km from N to S, averaging about 1.5 km wide. It is roughly crescent-shaped, with a coral reef and lagoon occupying the concave side of the crescent (Figs. 2, 3). The island is dominated by the spectacular mountains of Gower (875 m) and Lidgbird (777 m) which occupy the southern half (Fig. 4); elevations in the N do not exceed 210 m.

Brief reports on the geology of Lord Howe Island were given by Hill (1870), Wilkinson (1882), Etheridge (1889) and David (1889); more complete accounts were published by Standard (1963) and Game (1970). A brief summary of earlier work was provided by Sutherland & Ritchie (1974). All previous workers recognised that the island is dominantly volcanic, but that calcarenite of aeolian origin is of local importance. Standard (1963) and Game (1970) showed that a number of volcanic units could be dis-

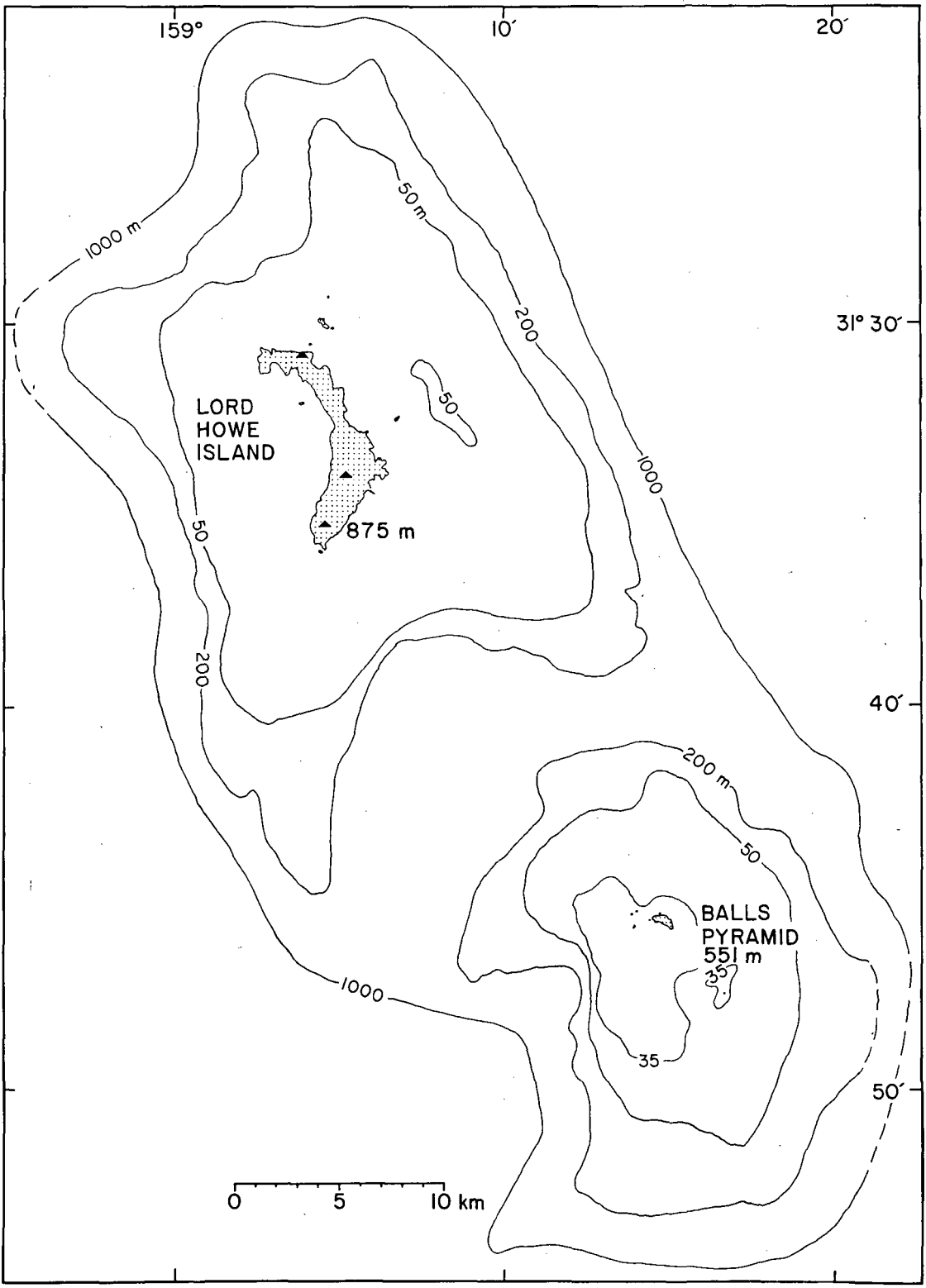


Fig. 2. Map showing bathymetry in the vicinity of Lord Howe Island and Balls Pyramid, after Royal Australian Navy Hydrographic Service (1965); isobaths in metres.

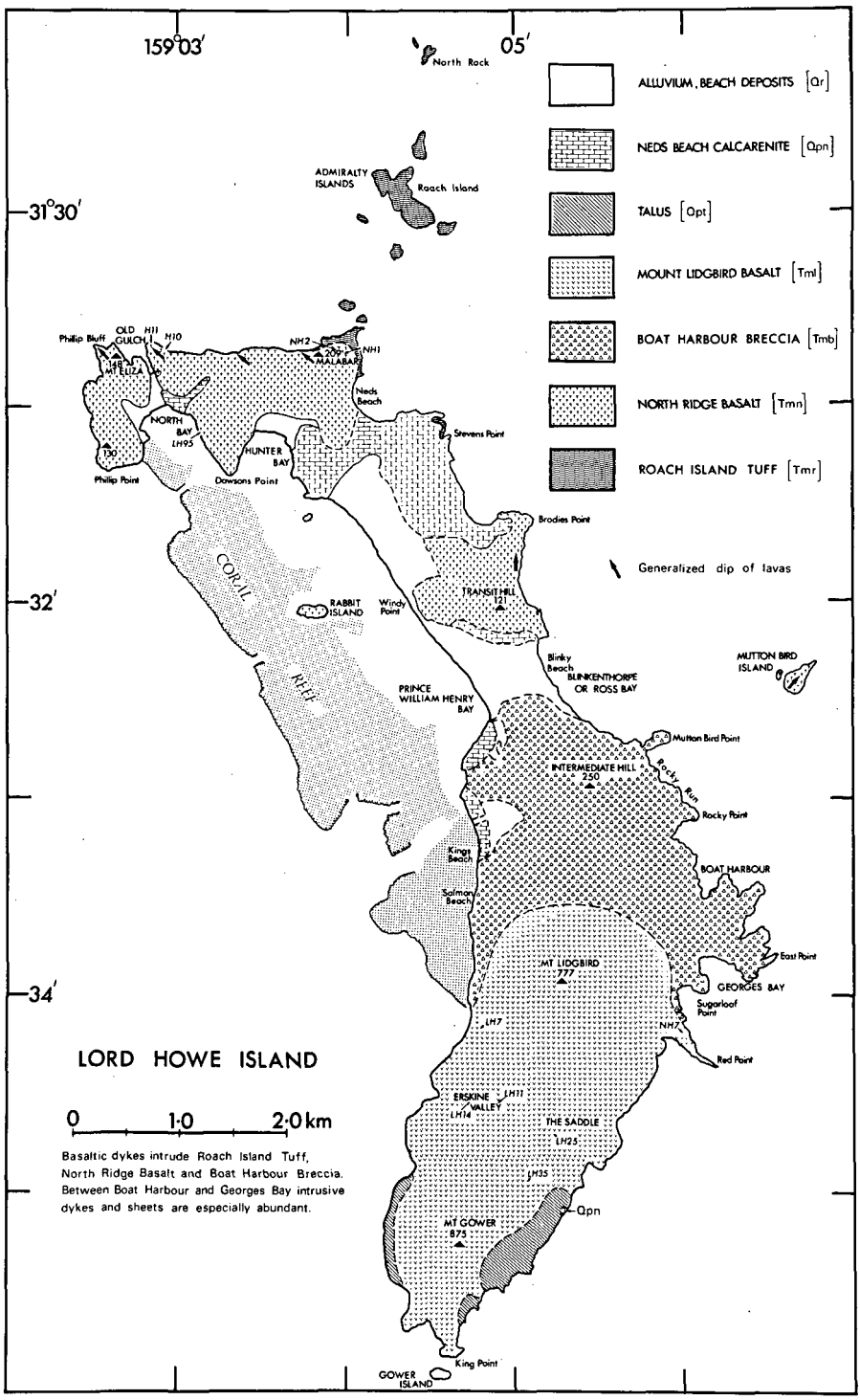


Fig. 3. Generalised geological map of Lord Howe Island. Location of samples used for geochronological measurements are indicated by sample number. Spot heights given in metres above sealevel. Dykes, which are common in the North Ridge Basalt and Boat Harbour Breccia, are not shown.

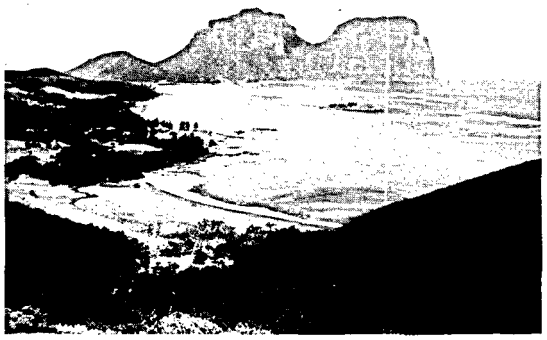


Fig. 4. View S from vicinity of Malabar on North Ridge, across the lagoon to Mount Gower (right), Mount Lidgbird (centre) and Intermediate Hill (centre left). Surf line at right indicates location of reef; Rabbit Island lies within lagoon. Caldera-filling, horizontal lavas (Mount Lidgbird Basalt), comprise Mount Gower and much of Mount Lidgbird. Margin of caldera runs through northern flanks of Mount Ldgbird.

tinguished, but they differed in their interpretations as to the nature and relative ages of these units.

In this paper we present results of a further geological study of Lord Howe Island based upon field mapping, isotopic age determinations and palaeomagnetic measurements. These data permit the origin and history of the island to be elucidated in some detail. We are able to demonstrate that Lord Howe Island is the eroded remnant of a much larger basaltic shield volcano that was active over a relatively short interval in the late Miocene.

GEOLOGY

A geological sketch map of Lord Howe Island, showing the distribution of the major units, is given in Figure 3. Following a brief outline of the geology and structure of the island, individual formations will be described in greater detail.

Apart from the locally erupted Roach Island Tuff, the oldest formation exposed on Lord Howe Island is the North Ridge Basalt, which underlies most of the northern half of the island. It consists of typical shield building basalt lavas that dip gently (~5°–8°) away from an eruptive centre located somewhere in the vicinity of Mount Lidgbird in the S of the island.

The Boat Harbour Breccia is a massive, well indurated, unbedded formation that occupies the country between the S end of Blinky Beach and the northern slopes of Mount Lidgbird, and is especially well exposed on the E coast. Its origin is uncertain, but it may be an agglomerate formed within the main throat of the volcano.

Both the North Ridge Basalt and Boat Harbour Breccia are intruded by numerous basaltic dykes, many of which strike toward Mount Lidgbird. Generally the dykes dip steeply toward the SW. On the E coast, especially S from Rocky Point, the proportion of dykes increases dramatically and the dips decrease to about 30° (to the SW) on average, to form a basaltic sheet complex, indicating proximity to a major vent.

The Boat Harbour Breccia and associated sheet complex are overlain by the Mount Lidgbird Basalt, which occupies the southern one-third of the island. Virtually the whole of the spectacular Gower-Lidgbird massif is composed of the Mount Lidgbird Basalt, an essentially flatlying sequence of lavas believed to be filling a large caldera formed by collapse of the summit of the shield volcano. This activity brought to a close the volcanic history of Lord Howe Island. Subsequently, erosion modified the volcano profoundly, so that only a small proportion of the original structure remains.

The youngest formation of note, apart from the coral reef, beach and alluvial deposits, is the Neds Beach Calcarenite, which is exposed over much of the country between Brodies Point and Neds Beach, and occurs elsewhere locally around the coast.

Roach Island Tuff

Among the oldest rocks exposed on Lord Howe Island are those comprising the Roach Island Tuff, which crops out in the northeasterly-trending point beneath Malabar, and in the Admiralty Islands (Fig. 3). Because of difficulties of access, most of the complex was viewed only from a boat. Although the name for this formation is derived from one of the Admiralty Islands, the best exposures occur on the N coast of Lord Howe Island beneath Malabar. Here the Roach Island Tuff extends to about 60 m above sealevel (Fig. 5), and consists of interbedded tuffs, thin rubbly lavas and beds of coarser fragmental material, with a dip of about 10° to the SW, indicative of an eruptive centre to the NE. On the N-trending section of coast, immediately E of Malabar, N of Neds Beach, indurated coarse breccia sheets, possibly talus or lahar material (as the basaltic rocks within them are commonly subangular to rounded), overlie weathered basaltic flows or spatter, and in turn are overlain by thin lava flows.

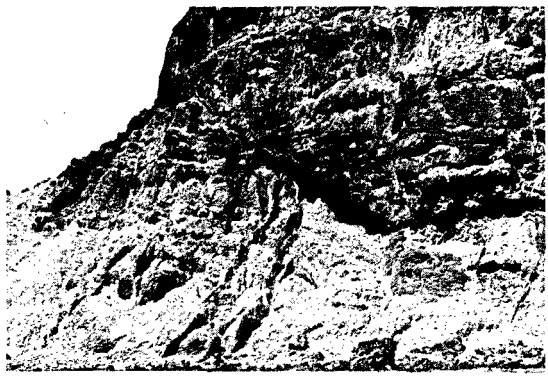


Fig. 5. Roach Island Tuff on N coast, overlain by North Ridge Basalt. View looking SSE, beneath Malabar. Tuffs, poorly cropping out in lower left, overlain by tuffs, fragmental material and thin lavas dipping to the SW. Contact with North Ridge Basalt nearly horizontal at lower right, extending diagonally across centre part of photograph. Dykes common in tuff complex, with some following along erosional disconformity and others intruding directly through into North Ridge Basalt.

Lavas of the North Ridge Basalt lap against and overlie the Roach Island Tuff. The contact is well exposed in the cliffs under Malabar (Fig. 5). Basaltic dykes cut through both the tuff complex and the North Ridge Basalt, and at least one dyke has intruded along the erosional disconformity between the two formations.

The Admiralty Islands are composed mainly of yellow tuffs, locally with dips of up to 25° or more. Basaltic dykes intruding the tuffs are particularly numerous, and their resistance to erosion no doubt accounts for the preservation of the Admiralty Islands.

A restricted outcrop, surrounded and overlain by younger Neds Beach Calcarenite, at Stevens Point, E of Neds Beach, consist of red tuffs and weathered, nearly horizontal lava flows, intruded by dykes. These tuffs and lava flows are regarded as part of the Roach Island Tuff.

The eruptions from the volcanic centre or centres, located just off the NE coast of Lord Howe Island, that produced the Roach Island Tuff, are thought to be subaerial. However, seawater probably had access to the vent or vents, accounting for the predominance of tuffaceous products, which perhaps indicates that sealevel was not very different then compared with the present.

North Ridge Basalt

The North Ridge Basalt underlies most of Lord Howe Island northward from its narrowest point at Blinky Beach, and crops out also on Mutton Bird Island. Standard (1963) named these rocks the North Ridge Volcanics, but as the dominant rock type is basalt, we have chosen to modify his terminology. North Ridge comprises the E-W trending high country between Malabar (209 m) and Mount Eliza (148 m), forming the northern part of Lord Howe Island. Game (1970) referred to the North Ridge Basalt as the 'horizontal altered series'.

The precipitous cliffs of the N face of North Ridge provide excellent exposures of the basalts, although most of the section is inaccessible. Part of this sequence can be reached, however, in the vicinity of Old Gulch. North Ridge is built of at least 30 lava flows, which have a consistent, approximately NW dip of the order of 5° to 8° (Fig. 6). The flows range from 1 m to 10 m thick, most commonly 2 m to 5 m thick, and typically have fragmental, often oxidised, tops and bottoms with more massive interiors. Individual lavas are seen to pinch out laterally. The basalts commonly are vesicular and locally amygdaloidal, with calcite especially abundant as a filling. Dykes, near vertical, and striking toward Mount Lidgbird, intrude the North Ridge Basalt in great abundance; Game (1970) reported the presence of at least 80 dykes in the cliffs of the N coast between Malabar and Old Gulch.



Fig. 6. Looking WSW across Old Gulch to Mount Eliza (148 m) on N coast, showing section through gently northwesterly dipping, shield building lavas (North Ridge Basalt).

The North Ridge Basalt has the appearance of a rapidly-erupted, shield building sequence of lavas, with no evidence of significant time breaks between successive lava flows. It should be noted, however, that disconformable relations are found locally. Thus at Phillip Point, lava flows intruded by dykes seem to be overlain with erosional disconformity by flows in which fewer dykes occur.

Weathered basalts, invaded by numerous dykes, crop out along the E coast from Brodies Point to Blinky Beach and dip gently to the N. There is little outcrop inland, but similar rocks are found at Windy Point on the W side of the island. Mutton Bird Island was only viewed from a boat, but it appears to consist of similar lava flows that dip about 5° approximately NE. Dykes are common. We are confident that the lavas of Mutton Bird Island and the Transit Hill area belong to the North Ridge Basalt.

The radial disposition of dips in the North Ridge Basalt about a focus in the vicinity of Mount Lidgbird is strong evidence that the lavas were erupted from a vent or rift zone located in that part of the island.

Lavas of the North Ridge Basalt typically have plagioclase phenocrysts with less common clinopyroxene and rather rare olivine, which is usually altered. The phenocrysts are set in an intergranular groundmass consisting of plagioclase, clinopyroxene, iron oxide and pale brown glass, commonly altered. Brown, yellow or green mineraloid is found in the groundmass of many basalts and calcite is common as a cavity infilling. The lavas range from relatively fresh to extensively altered rocks. They appear to have tholeiitic affinities; a brief discussion of their chemistry and comparisons with the compositions of the dykes and the Mount Lidgbird Basalt will be given in a later section.

Boat Harbour Breccia

Boat Harbour Breccia is well exposed on the E coast from the southern end of Blinky Beach to Red Point (Fig. 3). Particularly good and typical exposure occurs SW of Rocky Point, just N of Boat Harbour. Here, as elsewhere, the formation consists of massive, well-indurated breccia exhibiting no bedding, and composed of angular to subangular basaltic fragments ranging down in size from 1 m, but usually consisting of fragments less than 10 cm across, set in a fine-grained matrix (Fig. 7). The basaltic fragments range from aphyric to feldsparphyric types, some of which are vesicular or amygdaloidal; they show varying degrees of alteration. Such a range of rock types

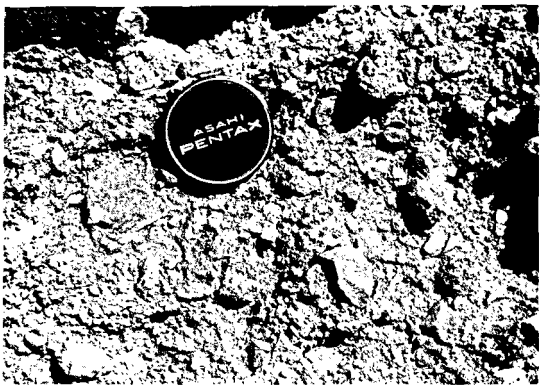


Fig. 7. Typical Boat Harbour Breccia on E coast, 300 m SW of Rocky Point. Lens cap 5.2 cm in diameter. Angular to subangular basaltic clasts range down from about 4 cm in size.

may be found within a single outcrop. However, especially in small outcrops in which the basaltic clasts appear to be of very similar composition to one another, the breccia is difficult to distinguish from a fragmental lava flow, although nowhere was a flow boundary observed.

Southward from Boat Harbour, only remnants of the breccia are found as screens between basaltic dykes and sheets. Locally good outcrops of Boat Harbour Breccia occur on the NE and N slopes of Intermediate Hill, and on the W coast, especially at the southern end of Salmon Beach. Virtually no outcrops are found on the higher parts of Intermediate Hill, above about 100 m altitude, but it is inferred from the distribution of outcrops of the breccia that it underlies the whole of this area, as shown on the map (Fig. 3). Nevertheless, a number of massive smooth outcrops of weathered, structureless, basaltic rock that are not breccia occur on the ridge running from Rocky Point toward the summit of Intermediate Hill, especially at an altitude of about 100 m. Similar outcrops are found at an altitude of about 60 to 80 m on the NE slopes of Intermediate Hill above Rocky Run. In thin section these rocks are seen to be extensively altered with development of chlorite, epidote and calcite, although clearly originally of basaltic composition. No lava flow tops or bottoms were found, and thus possibly the rocks may be intrusive bodies.

The origin of the Boat Harbour Breccia is problematical, but we suggest that it was formed mainly by explosive volcanism within the central vent region of the volcano, and therefore may be an agglomerate.

No contact was observed between the Boat Harbour Breccia and North Ridge Basalt, but it must lie in the waist of the island between Blinky Beach and Prince William Henry Bay. Thus it is not possible to determine whether the Boat Harbour Breccia is older than the North Ridge Basalt, as suggested by Game (1970), or younger. The paucity of palagonitic tuffs within the Boat Harbour Breccia favours the view that the breccia formed by explosive activity within the vent of an already large shield volcano. The presence of such a shield volcano would effectively limit the interaction of seawater with magma, thus providing an explanation for the relative lack of palagonitic tuffs. If this were the case, then the Boat Harbour Breccia would be younger than the shield building lavas comprising the North Ridge Basalt.

Relevant to this discussion is the occurrence of moderately steeply dipping, yellow to green tuffs containing lapilli of basalt glass and basaltic blocks, up to 0.5 m across, on the small peninsula forming Mutton Bird Point on the E coast. Locally, on the eastern tip of Mutton Bird Point, within a coarse breccia matrix, are found a number of pillow-like basalt bodies, up to 1 m in diameter, showing well-developed, glassy selvedges 1 to 2 cm thick and a crude radial joint pattern. These bodies are interpreted as parentless pillows brought up from deeper (submarine) levels by the explosive eruptions. The complex at Mutton Bird Point probably was produced as volcanism breached sealevel, prior to construction of the large shield volcano by the North Ridge Basalt. It appears to be similar in character, and possibly also age, to the Roach Island Tuff.

The great abundance of basaltic dykes and sheets in both the tuff complex at Mutton Bird Point and the Boat Harbour Breccia on the immediately adjacent coast, does not allow the relative ages of these two units to be established. If the tuff complex and the Rocky Point Breccia were formed at the same time, this would suggest that the latter was emplaced prior to the shield building phase, that is before the eruption of the North Ridge Basalt.

Thus, although we think that the Boat Harbour Breccia was emplaced within the vent regions of the shield volcano after it was built by the North Ridge Basalt, it is emphasised that the evidence remains equivocal.

Dykes and sheets

Basaltic dykes are found in great abundance intruding Roach Island Tuff, North Ridge Basalt and Boat Harbour Breccia. The dykes average 1 to 2 m in width, but range up to about 4 m wide. Many of the dykes, especially those in the northern half of the island, strike toward, or a little to the NE of Mount Lidgbird. The average strike of the 32 dykes and sheets from which samples were collected for palaeomagnetic measurement is 335° ± 33° (true), where the measure of dispersion is the standard deviation of the population. The dykes generally have steep dips (60°–90°) toward the SW, but shallower dips are not uncommon, and on the coast immediately N of Blinky Beach the dykes commonly dip toward the NE. Locally, as on Rabbit Island in the lagoon, dykes are so closely spaced that little or no country rock remains between them, and dykes commonly are found intruding earlier ones.

On the E coast, S of Mutton Bird Point, the basaltic intrusives in the Boat Harbour Breccia comprise about 30 percent of the outcrop. Their strike is dominantly NNW but commonly they have a much shallower dip, generally between about 20° and 40° to the SW, and thus may be regarded as sheets rather than dykes. These sheets range from about 0.3 to 2 m in thickness, averaging about 0.5 m. Southward from Rocky Point the proportion of sheets increases, such that at Boat Harbour it exceeds 80 percent of the

outcrop, with but small irregular screens of Boat Harbour Breccia between the sheets (Fig. 8). The inaccessible cliffs around Georges Bay, to the S of Boat Harbour, appear to consist almost entirely of closely-spaced sheets, dipping SW under Mount Lidgbird. Along the coast between Rocky Point and Boat Harbour a set of near vertical dykes also occur, with a southwesterly strike toward Mount Lidgbird, cutting across the basaltic sheet complex almost at right angles.

It is inferred that the dykes throughout the northern part of the island and the sheets in the Boat Harbour-Georges Bay area all belong to a single intrusive episode. They probably originated from the top of a magma chamber lying under the E flanks of Mount Lidgbird, and the sheets are interpreted as a cone sheet complex. The dykes intruding the cone sheet complex are probably only a little younger.

The intrusive dykes and sheets range from aphyric to plagioclasephyric, less commonly with clinopyroxene and rarely olivine as phenocrysts. The groundmass consists of plagioclase, clinopyroxene and iron oxide, with variable proportions of poorly crystallised mesostasis, commonly showing much evidence of hydrothermal and deuteric alteration. Especially in the Boat Harbour area, the intrusives are intensively altered to chlorite, carbonate and zeolite, probably because of proximity to the throat of the volcano. Margins of the dykes show evidence of chilling against the country rock or other intrusive bodies, and may also be vesicular and amygdaloidal.

Game (1970) used the term 'inclined altered series' to describe the rocks that occur on the E coast between Mutton Bird Point and Red Point. Though he recognised the presence of breccias and the intrusive basaltic dykes and sheets in this area, clearly he interpreted the shallow-dipping sheets exposed on the shores of Boat Harbour as lava flows, and invoked tectonic tilting to account for their attitude. The field evidence is sufficiently strong to discount Game's interpretation, and the palaeomagnetic data, to be discussed below, provide confirmation that the intrusive sheets have not been disturbed tectonically subsequent to their emplacement. This erroneous interpretation by Game (1970) also led him to suggest that these basaltic sheets were much older than the North Ridge Basalt, a conclusion we believe to be incorrect.

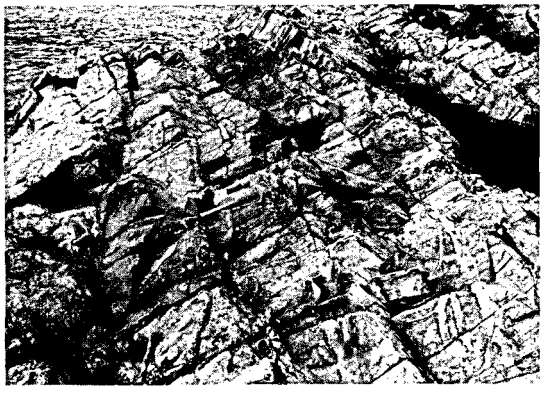


Fig. 8. Closely spaced basaltic sheets striking NNW and dipping 30° to 45° WSW, as exposed on W side of Boat Harbour. Hammer (33 cm long) lies on one sheet, with handle end abutting against another sheet. Little if any country rock (Boat Harbour Breccia) occurs between the basaltic cone sheets at this locality.



Fig. 9. Caldera-filling, horizontal and commonly columnarly jointed lavas of Mount Lidgbird Basalt in Mount Lidgbird (left) and Mount Gower (right). Caldera margin lies in vegetated slopes of Mount Lidgbird, passing through prominent vegetated buttress at centre of photograph about one third of the apparent height to top of Mount Lidgbird, just to the right and below light coloured vertical face in basalt. In lagoon at right occur eroded outcrops of Neds Beach Calcarenite.

Mount Lidgbird Basalt

The southern one third of Lord Howe Island comprises the Mount Lidgbird-Mount Gower massif, with its spectacular cliffs and precipices, rising to an altitude of 875 m at Mount Gower (Fig. 4). It is built almost entirely of essentially flatlying lavas, here termed the Mount Lidgbird Basalt, a slight modification to the name Mount Lidgbird Volcanics given by Standard (1963), in recognition of the basaltic nature of the flows. Game (1970) used the term Gower-Lidgbird series for this sequence of lavas.

The basalts range from tabular lavas, up to 30 m thick, some with well-developed columnar jointing, to thin flow units a few metres thick (Figs. 9, 10). Partly fragmental flows are present but are of minor importance. Interbedded tuffs appear to be rare. Oxidised tops of lava flows are common, but there is no evidence for extended intervals of time between successive flows.

It is clear that the extensive dyke and sheet swarm that intrudes the North Ridge Basalt and Boat Harbour Breccia is earlier than the Mount Lidgbird Basalt, a fact recognised by Game (1970). A few dykes, however, are found cutting the Mount Lidgbird Basalt in the cliffs on the W flanks of Mount Gower and on the ridge leading up from The Saddle to Mount Gower; these dykes no doubt are related to the same volcanic episode as that which formed the Mount Lidgbird Basalt.

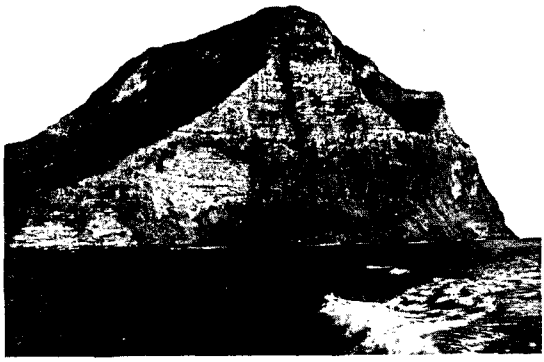


Fig. 10. W face of Mount Gower, showing horizontal lavas of caldera-filling sequence. Lower vegetated slopes at centre of photograph consist of talus.

The Mount Lidgbird Basalt is interpreted as a sequence erupted rapidly into a large collapse caldera that formed in the summit region of the shield volcano. The caldera appears to have been elliptical in plan with a major axis (NNE-SSW) of at least 5 km and a minor axis of at least 2 km (Fig. 3). These dimensions are minimum estimates, as erosion has removed the caldera margins on three sides.

The caldera margin is well exposed on the E coast on the northeastern flanks of Red Point peninsula (Fig. 3), where a sharp, if irregular, contact is observed between the cone sheet complex and the younger caldera-filling rocks. Red Point itself lies just within the caldera, and is built in part of tuffs and irregular basaltic bodies, presumably the result of local volcanic activity along the caldera boundary fault. The volcanics of Red Point are overlain by a conglomerate, up to about 40 m thick, dipping approximately 25° to the SW, probably a talus deposit originating from the caldera wall. In turn this conglomerate is overlapped by the flatlying lavas of the Mount Lidgbird Basalt.

On the northern flanks of Mount Lidgbird the caldera margin can be located only approximately, owing to lack of outcrop. However, in a prominent buttress on the NW flanks of Mount Lidgbird (Fig. 9), at an elevation of about 150 m, caldera-filling lavas and interbedded conglomerate lap onto a complex of basaltic dykes and fragmental material, possibly erupted along or adjacent to the caldera boundary fault, as at Red Point.

Game (1970) has given detailed petrographic descriptions of the Mount Lidgbird Basalt, and he presented much data on their chemistry and mineralogy. The basalts range from strongly porphyritic types to almost aphyric types. Moderately phyric lavas are most common with olivine as phenocrysts, but plagioclase and clinopyroxene occur also. The rocks typically have an intergranular groundmass, commonly exhibiting flow texture, and consisting of plagioclase, clinopyroxene, olivine (commonly iddingsitised), iron oxide and varying proportions (up to 25 percent) of mesostasis, which is poorly crystallised to glassy. The mesostasis is turbid with alteration and is partly replaced in some rocks by zeolite and carbonate, which also are found in vesicles. Some of the lavas are virtually holocrystalline, showing but minor deuteric alteration. The lavas are mildly alkaline basalts trending toward hawaiite; this will be treated in more detail below.

Neds Beach Calcerenite

As noted by previous workers, calcarenite is the most widespread sedimentary rock found on Lord Howe Island. We follow Standard (1963) in calling it the Neds Beach Calcarenite. It is best developed between Brodies Point and Neds Beach (Fig. 3), but calcarenite also occurs locally at the head of North Bay and along the W coast as far S as Kings Beach. What appears to be calcarenite, at least as viewed from a boat, crops out over about 100 m on the coast to the E of Mount Gower, overlying a large talus slope deposit (Fig. 3). From Neds Beach SE to Brodies Point there is virtually continuous outcrop of calcarenite, where it is found at least 45 m above sealevel. On the NE side of North Bay a tongue of calcarenite was deposited in a valley cut into North Ridge Basalt, to an altitude of 100 m. Twin lateral drainage has developed along the contacts between the calcarenite and basalt.

Standard (1963) has described the Neds Beach Calcarenite in some detail, so that only a brief description is necessary here. The calcarenite is nearly pure calcium carbonate, consisting mainly of fragments of coralline algae with less abundant coral, foraminifera and molluscs. It shows spectacular cross-bedding, and this, together with its distribution, leaves little doubt that it is aeolian in origin, as was recognised by the earliest observers. We agree with Standard (1963) that the calcarenite was formed during periods when sealevel was substantially lower than the present, presumably brought about by eustatic sealevel changes during glacial intervals in the Pleistocene and possibly late Pliocene. Lowering of sealevel would have resulted in an extensive area of the bevelled summit of the underlying seamount being exposed (Fig. 2), from which calcareous material was entrained by the winds and deposited on the slopes of the existing hills of Lord Howe Island.

Other formations

Extensive, vegetated talus deposits occur beneath the precipitous cliffs of Mount Gower (Fig. 3). That on the SE coast appears to extend to an altitude of the order of 200 m.

The existing coral reef and the lagoon on the W side of the island has been studied by Standard (1963). The present beach deposits, including the dune backing Blinky Beach, and the young alluvial deposits (Fig. 3) call for little additional comment.

Balls Pyramid

Balls Pyramid is a spectacular spire of volcanic rock rising to an altitude of 551 m, located 25 km SSE of Lord Howe Island. The islet is only 1100 m long (WNW-ESE) and about 400 m across at sealevel. It is separated from Lord Howe Island by water depths of the order of 500 m, and forms a separate culmination on the same broad, submarine base as that on which Lord Howe Island is built (Fig. 2).

Balls Pyramid was inspected from a boat; no landing was attempted. It is built predominantly of lavas, undoubtedly basaltic, which show a consistent dip of about 8° approximately to the NE (Fig. 11). In excess of 100 lava flows are visible on the northwestern face of Balls Pyramid, with individual lavas commonly only a few metres thick. Dykes are rare. Locally tuffs occur intercalated with the lavas, for example at a very approximate altitude of 180 m on the western ridge of Balls Pyramid. We interpret the lavas as a typical shield-building sequence, erupted from a centre located to the SW. This is consistent with the bathymetry (Fig. 2), as Balls Pyramid lies near the northeastern extremity of a submarine platform averaging 30 to 35 m below sealevel, thought to delineate approximately the outline of a former subaerial shield volcano.

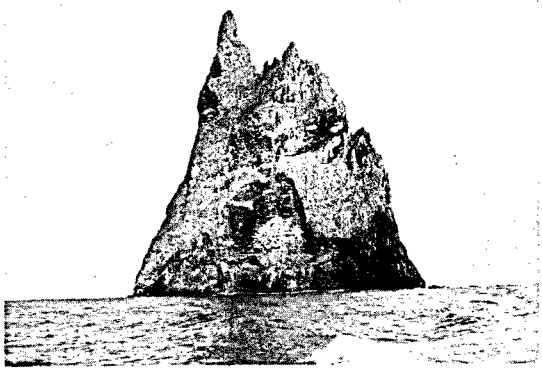


Fig. 11. Balls Pyramid (551 m) viewed from the NW. Thin shield-building lavas dip gently to the NE, extending to sealevel at left Lavas overlie, with irregular contact, a tuff complex intruded by dykes, which can be seen in the lower one eighth of the apparent height of Balls Pyramid. Steeply dipping tuff intercalated with lavas occurs on skyline, at approximately half the apparent height of the Pyramid. Vertical, light coloured streaking is mainly guano.

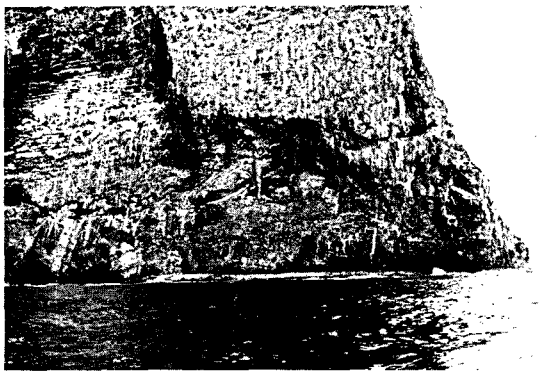


Fig. 12. Northwestern tip of Balls Pyramid viewed from N, showing basalts overlying tuffs intruded by dykes. Tuff with variable and moderate dips occupies one third of face at left and extends half way up face at centre of photograph. Note guano streaking, especially on basalts.

The shield-building lavas have flowed over and around a sequence of red, purple, grey and yellow tuffs and other fragmental material intruded by numerous dykes, that is well exposed from sealevel to a maximum altitude of perhaps 70 m around much of the base of Balls Pyramid (Fig. 12). However, the shield-building lavas extend to sealevel over a few hundred metres on the northeastern coast of Balls Pyramid.

Thus Balls Pyramid is the erosional remnant of a subaerial shield volcano, probably originally at least 6 km in diameter at sealevel, and possibly substantially larger. We can only surmise as to the age of this volcano, but suggest it was active at about the same time as that which constructed Lord Howe Island.

GEOCHEMISTRY

Major-element analyses were carried out on the five samples of lava from the North Ridge Basalt that were used for K-Ar age measurements. The results are given in Table 1, from which it can be seen that the lavas are typical basalts showing a restricted range of composition and characterised by relatively low MgO contents. Calculations show that each of the rocks is quartz normative; even when recalculated with all Fe as FeO, three of the five remain quartz normative and the average has but 0.1 percent olivine. Clearly the North Ridge Basalt is tholeiitic in character, as confirmed by the fact that the analyses plot in the tholeiite field on an alkalis–SiO_2 diagram (Irvine & Baragar, 1971).

Game (1970) presented analyses of five dykes; four intrude the North Ridge Basalt, the fifth probably intrudes the Boat Harbour Breccia. He also reported analyses for nine lavas from the Mount Lidgbird Basalt. Averages for these two sets of data, recalculated free of volatiles, are listed in Table 1. Because two of the analyses of Mount Lidgbird Basalt are on samples from lavas that have cumulative olivine, a second average analysis is given which excludes these two analyses. The standard deviations listed for each oxide in the average analyses show that the variability in composition generally is greater than for the North Ridge Basalt, but that it is still relatively small.

Overall the chemistry of the dykes and the Mount Lidgbird Basalt is similar to that of the North Ridge Basalt. Nevertheless, important differences are found. The younger rocks have slightly lower SiO_2, slightly higher MgO, significantly lower CaO, and significantly higher alkalis compared with the North Ridge Basalt. These differences are reflected in the norm calculations, in that the dykes and Mount Lidgbird Basalt are olivine normative, and the latter is nepheline normative when recalculated with all Fe as FeO. The average analyses plot within the alkali basalt field on the alkali-SiO_2 diagram of Irvine & Baragar (1971). The dykes are olivine basalts with characteristics transitional between tholeiitic and alkali basalt, whereas the Mount Lidgbird Basalt appears to be alkali basalt tending toward hawaiite. Some of the individual analyses from the Mount Lidgbird

TABLE 1. Chemical analyses and CIPW norms of North Ridge Basalt, and average analyses of dykes and Mount Lidgbird Basalt.

	NORTH RIDGE BASALT					AVERAGE ANALYSES			
	H10 Old Gulch	H11 Old Gulch	NH1 Malabar	NH2 Malabar	LH95 North Bay	North Ridge Basalt (5)	Dykes (5)	Mount Lidgbird Basalt (9)	Mount Lidgbird Basalt (7)	
SiO_2	50.91	51.47	49.79	49.81	50.05	51.3 ± 0.7	49.2 ± 1.7	48.5 ± 2.0	49.3 ± 1.3	SiO_2
TiO_2	2.52	2.66	2.47	2.72	2.56	2.6 ± 0.1	2.6 ± 0.2	2.7 ± 0.5	2.8 ± 0.4	TiO_2
Al_2O_3	14.58	14.89	14.77	14.74	15.25	15.1 ± 0.3	14.8 ± 1.1	15.1 ± 2.1	15.9 ± 1.4	Al_2O_3
Fe_2O_3	2.88	4.64	3.34	2.97	2.71	3.4 ± 0.8	3.2 ± 1.8	4.4 ± 1.5	4.4 ± 1.7	Fe_2O_3
FeO	7.54	5.46	6.95	7.92	7.06	7.1 ± 1.0	7.9 ± 1.7	7.5 ± 1.5	7.2 ± 1.2	FeO
MnO	0.18	0.17	0.18	0.18	0.15	0.17 ± 0.01	0.17 ± 0.02	0.15 ± 0.02	0.15 ± 0.02	MnO
MgO	4.90	4.58	5.61	5.77	5.55	5.4 ± 0.5	7.8 ± 3.2	7.9 ± 4.4	6.0 ± 1.8	MgO
CaO	10.56	10.27	10.98	10.31	10.52	10.7 ± 0.3	9.4 ± 1.1	7.9 ± 1.7	7.7 ± 1.9	CaO
Na_2O	2.78	3.06	2.85	2.83	2.94	2.9 ± 0.1	3.3 ± 0.8	3.6 ± 1.2	4.0 ± 0.9	Na_2O
K_2O	0.93	0.91	0.62	0.79	0.80	0.8 ± 0.1	1.0 ± 0.2	1.5 ± 0.5	1.7 ± 0.3	K_2O
P_2O_5	0.40	0.39	0.37	0.39	0.36	0.39 ± 0.02	0.39 ± 0.05	0.53 ± 0.17	0.57 ± 0.15	P_2O_5
Loss	1.99	1.57	1.66	1.99	1.91	-	-	-	-	Loss
TOTAL	100.17	100.07	99.59	100.42	99.86					TOTAL

	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)	
ne	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2.0	-	2.4	ne
qz	4.4	1.1	6.8	1.5	2.9	-	2.2	-	2.2	-	3.7	-	-	-	-	-	-	-	qz
or	5.6	5.6	5.5	5.5	3.7	3.8	4.7	4.8	4.8	4.8	4.8	4.8	5.9	5.9	8.9	8.9	10.1	10.1	or
ab	23.9	24.0	26.3	26.4	24.6	24.7	24.3	24.4	25.4	25.5	24.9	24.9	27.9	27.9	30.4	26.7	33.9	29.5	ab
an	25.0	25.1	24.6	24.6	26.2	26.3	25.6	25.7	26.6	26.7	25.6	25.6	22.6	22.6	20.6	20.6	20.4	20.4	an
di	20.9	21.5	19.5	20.5	21.7	22.4	19.3	19.8	19.7	20.1	20.2	20.7	17.3	17.6	12.1	12.4	11.3	11.7	di
hy	10.0	16.9	4.5	15.3	10.2	14.5	13.3	15.8	11.5	14.4	9.9	17.5	6.6	0.1	2.7	-	1.7	-	hy
ol	-	-	-	-	-	2.7	-	3.4	-	2.6	-	0.1	8.9	19.4	12.3	22.4	9.4	18.8	ol
mt	4.2	-	6.8	-	4.9	-	4.4	-	4.0	-	4.9	-	4.6	-	6.4	-	6.4	-	mt
il	4.9	4.9	5.1	5.2	4.8	4.8	5.3	5.3	5.0	5.0	5.0	5.0	4.9	4.9	5.1	5.1	5.3	5.3	il
ap	1.0	1.0	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.3	1.3	1.3	1.3	ap

Notes: Analyses of North Ridge Basalt by P. Beasley, mainly using XRF methods.
CIPW norm (a) calculated from analysis as reported, (b) calculated with all Fe as FeO.
Average analyses for dykes and Mount Lidgbird Basalt derived from Game (1970); number of analyses given in brackets.
The average analysis for Mount Lidgbird Basalt using 7 analyses omits analyses 2 and 3 of Game (1970), Table VI.

Basalt, however, are hypersthene normative, and may also be classified as transitional.

Thus the main shield-building lavas (North Ridge Basalt) are tholeiitic basalts, the dykes are olivine basalts, and the caldera-filling lavas (Mount Lidgbird Basalt) are transitional to alkali olivine basalts and hawaiites. This change in composition with time from tholeiitic to alkaline is commonly observed in oceanic volcanoes, for example in many of the Hawaiian volcanoes (Macdonald & Katsura, 1964) and in Réunion (Upton & Wadsworth, 1972).

GEOCHRONOLOGY

Methods

Samples were collected from many of the lavas and dykes for possible potassium-argon (K-Ar) age determination. Each sample was examined in thin section, but very few were holocrystalline and free of alteration. Nevertheless, some samples met the minimum criteria for dating as whole rocks. Each sample was crushed to a fragment size of -1 mm $+$ 0.5 mm, an aliquant was taken and crushed to less than 0.15 mm, and this representative sample was reserved for potassium analysis by flame photometry. The coarser material was used for analysis of argon by fusing a weighed amount, usually 5 to 20 g, in a high vacuum extraction line, mixing the evolved gases with an ^{38}Ar tracer; following purification, the argon was isotopically analysed in an AEI MS-10 mass spectrometer, as described previously (McDougall & Schmincke, 1977). All data were taken digitally on-line with a computer, in which all data reduction was done. To minimise possible isotopic fractionation of loosely-held atmospheric Ar in alteration products, the samples were not heated above 100°C in the vacuum system prior to Ar extraction. The calculated ages are reported in terms of the ^{40}K decay constants recently recommended by Steiger & Jäger (1977).

Results

Dating was carried out on 12 whole rock samples of basalt, five of which were from the North Ridge Basalt and seven from the Mount Lidgbird Basalt. No samples from dykes or sheets were regarded as sufficiently fresh for dating purposes.

Results are given in Table 2, in stratigraphic order. All K and Ar determinations were made in duplicate. In general the K results agree to better than one percent, and with few exceptions the duplicate measurements of radiogenic Ar agree to within two percent. The uncertainties quoted for an individual age are at the level of one standard deviation, derived by quadratically combining the uncertainties in the K and Ar measurements and the calibration of the ^{38}Ar tracer (McDougall et al., 1969). An average age is reported for each sample, and its reported uncertainty is either the standard deviation of the replicates or the standard deviation of the best individual result, whichever is the larger. This is done because in those cases where very good agreement is obtained between duplicates, the standard deviation of the average is unrealistically low, as there are combined uncertainties of about one percent in the calibration of the ^{38}Ar tracer and in the K analyses.

North Ridge Basalt

The calculated ages for the five samples from the North Ridge Basalt range from 6.62 $\pm$ 0.22 Ma to 7.20 $\pm$ 0.08 Ma (Table 2), forming a relatively concordant set of data which yield an overall mean of 6.92 $\pm$ 0.22 Ma. The dated samples were all collected from the sequence exposed in North Ridge, and probably come from lavas separated by not much more than 100 m stratigraphically. From the geology it might have been expected that all the ages would agree to within the uncertainties, but this is not the case. The oldest measured age was obtained from a sample on Malabar, probably the stratigraphically youngest lava dated. Thus it is apparent that the rocks have not behaved as perfect K-Ar systems, probably because of some loss of radiogenic argon by diffusion. The samples were the best available, but all show some deuteric alteration and contain minor amounts of mineraloid. Sample H11 has a few percent of apparently fresh, pale brown glass in the groundmass, and yields the lowest measured age. Similarly H10 contains about 5 percent of pale brown, isotropic glass, but the other samples appear to be virtually holocrystalline. The fact that less than 10 percent of the argon from samples H10 and H11 is radiogenic also indicates that they must be somewhat altered rocks.

Though the slightly discordant data led us to conclude that the ages are minimum estimates of the time of eruption and cooling of the lavas, we think that the mean age of 6.9 $\pm$ 0.2 Ma is unlikely to be more than several percent too young. This age is late Miocene, as the Miocene-Pliocene boundary has an estimated age of about 5.1 Ma (Berggren & van Couvering, 1974).

Mount Lidgbird Basalt

Of the large number of samples collected from the Mount Lidgbird Basalt, only seven were sufficiently well crystallised and fresh to be dated. The measured K-Ar ages range from 4.40 $\pm$ 0.13 Ma to 6.42 $\pm$ 0.14 Ma, but if the result from sample LH7 is omitted, the range is reduced substantially and becomes 5.85 $\pm$ 0.06 Ma to 6.42 $\pm$ 0.14 Ma, yielding a mean age of 6.26 $\pm$ 0.22 Ma. As with the North Ridge Basalt, the spread in apparent age is somewhat greater than might be expected from the geology, and no systematic variation in measured age is found with stratigraphic position.

The four samples with ages agreeing at 6.38 $\pm$ 0.03 Ma (NH7, LH14, LH3, LH25) are all essentially holocrystalline and contain less than one percent of intersertal mesostasis or glass. The remaining three samples, especially LH7, which gives the lowest age, contain a few percent of altered mesostasis, and small amounts of zeolite were identified in each of them. It is suggested that variable loss of radiogenic argon has occurred from these rocks. The excellent concordance of four of the ages from samples collected over 300 m of section, is taken to indicate that the Mount Lidgbird Basalt was erupted over a short time interval at 6.4 $\pm$ 0.1 Ma ago, in the late Miocene.

Game (1970) reported a K-Ar age of 7.7 $\pm$ 0.2 Ma for a plagioclase separate from a sample of Mount Lidgbird Basalt, significantly older than the

TABLE 2. Potassium-argon ages on whole rock samples of basalt from Lord Howe Island.

Field No.	Lab. No.	K (wt.%)	Rad. ^{40}Ar (10^{-11}mol/g)	100 Rad. ^{40}Ar / Total ^{40}Ar	Calculated Age (Ma) ± 1s.d.	Average Age (Ma) ± 1s.d.	Altitude (m)	Locality
Mount Lidgbird Basalt								
LH 35	73-630	1.002,1.003	1.067 1.075	90.2 85.4	6.13 ± 0.07 6.17 ± 0.05	6.15 ± 0.05	685	Mount Gower
LH 25	73-620	0.989,0.987	1.080 1.102 1.127	66.8 69.0 69.0	6.29 ± 0.05 6.42 ± 0.08 6.56 ± 0.06	6.42 ± 0.14	405	Mount Gower
LH 11	73-606	1.071,1.076	1.095 1.085	68.8 67.8	5.87 ± 0.07 5.82 ± 0.06	5.85 ± 0.06	150	Erskine Valley
LH 3	73-598	1.024,1.024	1.140 1.141	64.6 76.5	6.40 ± 0.08 6.41 ± 0.08	6.40 ± 0.08	120	SW flanks Mount Lidgbird
LH 14	73-609	1.097,1.094	1.210 1.214	69.2 66.7	6.35 ± 0.08 6.38 ± 0.06	6.36 ± 0.06	100	Erskine Valley
LH 7	73-602	1.027,1.027	0.801 0.768	72.2 75.6	4.49 ± 0.05 4.30 ± 0.05	4.40 ± 0.13	80	SW flanks Mount Lidgbird
NH 7	75-56	1.264,1.271	1.425 1.372	50.9 25.3	6.47 ± 0.06 6.23 ± 0.09	6.35 ± 0.17	90	Red Point
North Ridge Basalt								
NH 2	75-51	0.624,0.626	0.783 0.784	34.2 35.1	7.21 ± 0.08 7.22 ± 0.10	7.20 ± 0.08	150	Malabar
NH 1	75-50	0.529,0.528	0.617 0.634	30.5 41.1	6.71 ± 0.07 6.90 ± 0.07	6.80 ± 0.13	120	Malabar
H 11	73-663	0.748,0.747	0.875 0.844	8.1 4.4	6.74 ± 0.22 6.50 ± 0.39	6.62 ± 0.22	5	Old Gulch
H 10	73-662	0.750,0.750	0.904 0.903	9.4 9.3	6.94 ± 0.19 6.93 ± 0.20	6.93 ± 0.19	5	Old Gulch
LH 95	75-77	0.627,0.624	0.758 0.774	42.8 42.4	6.97 ± 0.07 7.12 ± 0.08	7.05 ± 0.11	5	North Bay

$\lambda_e + \lambda_{e'} = 0.581 \times 10^{-10} a^{-1}$ $\lambda_\beta = 4.962 \times 10^{-10} a^{-1}$ $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol Rad. ^{40}Ar = Radiogenic ^{40}Ar

TABLE 3. Palaeomagnetic data on lava flows from North Ridge Basalt.

Unit Number	N	K	α_{95}	D	I	Lat	Long
Headland just north of Blinky Beach							
LH79	4	248	5.8	34	-42	59	244
LH80	3	176	9.3	34	-38	58	239
LH81	2	-	-	31	-42	62	242
LH84	4	197	6.5	31	-35	59	233
Coast just east of Old Gulch							
LH86	4	829	3.1	6	-61	78	316
LH87	4	611	3.7	13	-65	71	310
LH89	4	66	11.3	348	-70	66	356
LH90	5	86	8.3	28	-71	59	309
North Beach to Dawsons Point							
LH94	4	160	7.2	12	-41	77	215
LH95	4	748	3.3	11	-49	81	243
LH97	4	435	4.4	13	-44	77	225
LH100	4	98	9.3	24	-54	69	263
LH101	3	1347	3.3	20	-58	72	278
LH108	5	494	3.4	34	-83	43	328
Malabar							
LH123	2	-	-	350	-42	78	104
LH124	3	59	16.3	355	-49	85	90
LH126	3	45	18.5	356	-45	84	118

N = number of unit vectors (drill cores) yielding the average D and I

K = estimate of precision parameter (Fisher, 1953)

α_{95} = semiangle of the cone of confidence about the mean, calculated at the 95 percent confidence level

D = declination in degrees, east positive

I = inclination in degrees, negative upwards

Lat = latitude of the virtual geomagnetic pole in degrees (positive-north; negative-south)

Long = longitude of the virtual geomagnetic pole, measured in degrees east of Greenwich

ages found in the present study. In the absence of information on how this age was measured, it is difficult to evaluate the reason why such differences exist.

Discussion

The K-Ar ages measured on the North Ridge Basalt and the Mount Lidgbird Basalt indicate that the subaerial part of Lord Howe Island was built over a relatively short period, in the order of 0.6 Ma. Thus Lord Howe Island volcano is very similar in this respect to many other volcanoes in oceanic regions, in that activity commonly is confined to intervals of less than 1 Ma. This is the case for volcanoes in Hawaii (McDougall, 1964) and many other volcanoes in the south Pacific (Duncan & McDougall, 1976; McDougall & Duncan, 1980).

We believe our data effectively discount the suggestion made by Game (1970) that subaerial volcanism on Lord Howe Island commenced in the mid-Tertiary, some 30 Ma ago.

PALAEOMAGNETISM

Methods

Samples were collected from most of the igneous rock units on Lord Howe Island by means of a portable, gasoline-powered drill, following techniques described by Collinson et al. (1967). The cores were oriented using a sun compass, by geographic sighting, or by magnetic compass. Normally a minimum of three cores was obtained from each lava, dyke or sheet sampled.

Laboratory procedures for measurement of the direction of magnetisation are similar to those described previously (McElhinny, 1973). Alternating field (AF) demagnetisation procedures were used on all samples, and in some cases thermal demagnetisation was undertaken; these techniques were employed to investigate the stability spectra of the magnetisation components. AF demagnetisation adequately isolated the primary magnetisation where only one thermoremanent component was present; these tests were carried out in incremental fields to a peak alternating field of 100 mT. Mean directions of magnetisation were calculated at each step, and the results listed in Tables 3 to 6 were selected on the basis of the demagnetisation level that gave the smallest scatter of directions. Where the magnetisation was clearly complex, vector analysis, in conjunction with thermal demagnetisation, was carried out. This procedure permitted elucidation of the complex magnetisation history of the Boat Harbour Breccia.

North Ridge Basalt

Fourteen lavas were sampled from the sequence in North Ridge at Malabar, on the coast just E of Old Gulch and on the eastern side of North Bay. A further five lavas were sampled from the headland just N of Blinky Beach. Results from cores obtained from two lavas indicated that the magnetisation was unstable and were not considered further.

All lavas collected from the North Ridge Basalt with stable magnetisation (17 sites) have normal polarity (Table 3). With few exceptions, the within-site scatter of directions is small; those sites with the α_{95} of directions greater than 10° perhaps reflect inherent instability of the magnetisation. All sites from North Ridge, collected over a stratigraphic thickness in the order of 100 m, yield a mean direction that is indistinguishable from the directions of the present geomagnetic field or the axial dipole field, and the calculated α_{95} is within the range expected from secular variation (Table 6; McElhinny & Merrill, 1975). The results from the locality just N of Blinky Beach are somewhat displaced from those from North Ridge, and may simply be recording another interval of the normal secular variation spectrum. When combined with the North Ridge data, the overall mean and the dispersion of the data about the mean still incorporate the present field, but show a 2° divergence from the axial dipole field direction (Fig. 13).

Boat Harbour Breccia

Boat Harbour Breccia was sampled on the coast just to the SW of Rocky Point (LH116, 118, 119), and on the W coast just S of Salmon Beach (LH102, 105, 107). In both localities younger dykes and sheets cut the breccia; the samples were collected from 1 to

TABLE 4. Palaeomagnetic data on dykes and sheets intrusive into North Ridge Basalt and Boat Harbour Breccia.

Unit Number	Attitude Strike	Dip	N	K	α_{95}	D	I	Lat	Long
Boat Harbour									
LH37	350	65SW	4	707	3.5	182	55	-86	134
LH38	340	50SW	4	213	6.3	203	50	-70	73
LH39	25	90	4	558	3.9	192	42	-77	36
LH40	335	35SW	4	930	3.0	187	49	-84	56
LH41	330	~30SW	4	53	12.7	180	52	-78	158
LH42	330	~35SW	4	302	5.3	164	51	-77	245
Brodies Point									
LH75	360	90	3	13	10.5	187	48	-87	2
LH77	360	90	4	246	5.8	178	56	-85	176
Headland just north of Blinky Beach									
LH82	350	40E	3	49	17.7	174	48	-84	274
LH83	340	70E	3	411	6.0	198	59	-73	102
LH85	295	45SW	4	49	13.2	173	47	-83	278
Coast just east of Old Gulch									
LH88	355	90	3	319	6.9	354	-40	80	127
LH91	335	70W	3	234	8.0	165	47	-76	259
LH92	340	80W	3	399	6.3	176	70	-68	165
LH93	335	90	4	133	7.9	175	54	-85	214
North Beach to Dawsons Point									
LH96	325	70SW	3	1028	3.8	185	71	-66	152
LH98	320	80SW	3	200	8.7	11	-47	79	234
LH99	325	80SW	3	606	5.0	159	34	-67	278
LH109	310	80SW	3	1709	2.9	5	-58	81	312
Rabbit Island									
LH110	325	65SW	3	2277	2.5	196	42	-74	44
LH111	315	75SW	3	397	6.1	175	30	-74	321
LH112	325	70SW	3	796	4.3	20	-43	71	235
Salmon Beach									
LH103	270	80S	3	206	8.6	158	60	-70	215
LH104	310	70S	2	-	-	175	56	-83	198
LH106	260	30S	3	75	14.3	163	51	-76	246
Just southwest of Rocky Point, Boat Harbour									
LH113	325	35SW	3	174	9.3	194	36	-72	31
LH114	35	75SW	4	163	7.2	187	53	-84	88
LH115	310	45SW	3	56	16.6	190	49	-81	60
LH117	35	80SE	3	58	16.3	166	47	-77	261
LH120	55	90	3	1007	3.8	193	51	-79	74
LH121	325	40SW	4	400	4.5	193	46	-77	52
LH122	325	35SW	3	157	9.8	182	28	-73	347

3 m from the nearest intrusive body. The NRM directions were scattered, and even following AF demagnetisation the scatter was not reduced appreciably, although there was an indication that the Rocky Point specimens were tending toward reversed magnetisation. Therefore, attempts were made to apply thermal demagnetisation techniques. Treatment at 10 steps from 100 to 650°C revealed a complex multicomponent magnetisation. At temperatures between 300 to 540°C, specimens from both localities exhibit dominant reversed magnetisation. Upon treament at higher temperatures a normal magnetisation component becomes dominant. The results from the thermal work are best represented by the difference vectors calculated after each demagnetisation step. Two examples, one from each locality, are shown in Figure 14. Below 200°C a normal component is removed, then a reversed component is consistently removed up to 540°C, and between 540 and 650°C a normal component is again removed (Table 6). The general

TABLE 5. Palaeomagnetic data on lava flows and a dyke of the Mount Lidgbird Basalt.

Unit Number	Altitude ft	Altitude m	N	K	α_{95}	D	T	Lat	Long
Southwest slopes, Mount Lidgbird									
LH1	560	170	6	581	2.8	3	-51	90	280
LH2	500	150	6	291	3.9	1	-51	89	261
LH3	410	125	5	891	2.6	3	-46	85	193
LH4	390	120	5	333	4.2	1	-47	86	176
LH5	320	100	5	1606	1.9	4	-56	84	310
LH6	280	85	6	321	3.7	8	-57	81	295
LH7	260	80	6	462	3.1	356	-56	84	12
LH8	410	125	5	23	16.4	20	52	73	258
Erskine Valley									
LH9	570	175	6	720	2.5	359	-57	84	343
LH10	540	165	6	1601	1.7	3	-51	88	253
LH11	490	150	6	818	2.3	4	-51	86	246
LH12	440	135	5	1238	2.2	11	-46	80	231
LH13	400	120	5	185	5.6	3	-50	87	229
LH14	330	100	5	520	3.4	9	-60	78	304
LH15	290	90	6	1006	2.1	0	-55	86	343
LH16	230	70	4	833	3.2	359	-55	86	352
LH17	140	45	6	804	2.4	359	-49	88	130
LH18	80	25	5	992	2.4	354	-48	85	93
LH19	5	0	5	988	2.4	15	-57	76	280
LH21	580	175	5	170	5.9	7	-54	84	277
LH22	630	190	5	301	4.4	3	-51	88	255
LH23	650	200	6	557	2.8	3	-54	86	301
The Saddle - Mount Gower Ridge									
LH24	1250	380	4	113	8.6	45	-43	50	252
LH25	1330	405	3	432	5.9	43	-54	54	266
LH26	1380	420	4	600	3.8	49	-59	50	276
LH27	1510	460	4	307	5.3	70	-75	37	305
LH28	1590	485	4	849	3.2	357	-29	74	149
LH29	1650	500	4	18	27	324	-52	62	34
LH30	1760	535	4	21	20	330	-42	63	77
LH31	1930	590	4	74	10.8	250	-66	11	19
LH32	2100	640	4	221	6.2	335	-52	69	60
LH33	2600	790	1	-	-	310	-16	38	85
LH34	2300	700	2	-	-	289	-16	20	72
LH35	2200	670	3	105	12.1	9	-50	82	248
*LH36	2100	640	2	-	-	336	-25	61	103

* Dyke

observation is that the breccia was magnetised over a period of time sufficiently long to encompass two polarity changes; however, the possibility that the low temperature component is a viscous remanent magnetisation cannot be ruled out. The higher temperature components, one normal and one reversed, were probably acquired during thermal events associated with the formation of the volcano. The hardest component (normal polarity) may date from the time of breccia formation, and the intermediate component (reversed polarity) may have been acquired contemporaneously with dyke and sheet injection. The low temperature overprint may have been acquired during the filling of the caldera by the normal polarity Mount Lidgbird Basalt. The demagnetisation data, although revealing a complex magnetisation history, are scattered and do not justify definition of averaged normal and reversed polarity components.

Dykes and sheets

Oriented samples were collected from each of 32 dykes and sheets, well-distributed through the North Ridge Basalt in the northern half of the island and in the Boat Harbour Breccia. Seven of the sites were in basaltic sheets (dips $< 45°$) on the western shores of Boat Harbour; although the sheets are considered to be just dykes with a shallower dip, results from them will be considered separately, at least initially.

With few exceptions, samples from individual intrusives yielded well-grouped directions subsequent to AF demagnetisation (Table 4). Only at one site does the within-site scatter exceed 18° at the 95 percent confidence level. Twenty-eight of the sites show reversed polarity and the other four have normal polarity, three of which are from North Ridge and the fourth from Rabbit Island in the lagoon.

TABLE 6. Mean palaeomagnetic data for volcanic units, Lord Howe Island.

Unit	N	K	α_{95}	D	I	Lat	Long
Mount Lidgbird Basalt							
Mount Gower, excluding dike	12	5.4	20.5	349	-56	77	43
SW slopes and Erskine Valley	22	242	1.9	3.5	-52.5	87	277
Dykes and Sheets							
Sheets (LH40,41,42,113, 115,121,122)	7	39	9.8	185	46	-84	28
All reversed polarity	28	40	4.3	180	50	-89	0
All normal polarity	4	54	12.5	8	-47	83	227
All sites	32	41	4.0	181	50	-89	30
**Boat Harbour Breccia*							
Salmon Beach NRM-200°	8	3.4	35	329	-55	-	-
400°-450°	8	5.3	26	171	51	-	-
600°-630°	8	3.1	38	297	-44	-	-
Rocky Point NRM-200°	7	32.7	11	37	-63	-	-
400°-450°	7	3.9	35	198	65	-	-
600°-630°	7	13.7	17	46	-70	-	-
North Ridge Basalt							
Just north of Blinky Beach	4	496	4.1	33	-39	60	239
All other sites	13	31	7.6	8	-57	81	294
All sites	17	24	7.4	15	-53	76	263
***Overall Means*							
All normal polarity	43	47	3.2	8.5	-52.4	82.6	263
All reversed polarity	28	40	4.3	180.4	50.3	-89.4	16.9
All units	71	42	2.6	5.2	-51.7	85.5	261.2

* Mean direction based on difference vectors calculated in conjunction with thermal demagnetisation

** Excluding data from Boat Harbour Breccia and the samples from the Saddle-Mount Gower Ridge section of the Mount Lidgbird Basalt

Note: All directions of magnetisation are given with respect to the present horizontal, i.e. dip of each unit is regarded as primary, so that no bedding corrections applied.

Analysis of the directions of magnetisation of intrusives from different geographical areas does not allow any distinction to be drawn between the several areas, so that all results have been pooled (Table 6). Note that the directions of magnetisation of the seven sheets from Boat Harbour are all reversed and concordant with those of the reversed polarity dykes. This provides strong support for the view that the sheets simply are dykes with a lower dip, and that the sheets are not tectonically tilted lavas as was suggested by Game (1970).

The overall mean direction for the reversed polarity intrusives has a low dispersion ($\alpha_{95} = 5.3°$) and is indistinguishable from the direction of the axial dipole field, but significantly different from the present geomagnetic field direction (after correction for polarity; Fig. 15). The low dispersion of the directions may be interpreted as indicating emplacement over an interval of time short in comparison with that required to sample the full range of directions expected on the basis of secular variation. The four normal polarity dykes yield a mean direction of magnetisation that is indistinguishable from that of the present field and the axial dipole field, and 180° from the mean direction of the reversed polarity dykes, within the uncertainties (Fig. 15).

The palaeomagnetic results from the basaltic dykes and sheets indicate that the majority were emplaced during a relatively short interval, when the earth's magnetic field had reversed polarity, and that some dykes were intruded during a period of normal polarity. In the absence of contact relations it is not possible to determine the relative ages of the normal and reversed polarity dykes.

Mount Lidgbird Basalt

Sampling of the Mount Lidgbird Basalt was carried out in three main areas. A group of 7 lavas was sampled in a cliff face just to the SW of Mount Lidgbird; 15 flows were collected from Erskine Valley, mostly within the stream bed, and 12 flows and one dyke were sampled on the ridge leading up from The Saddle toward Mount Gower (Fig. 3).

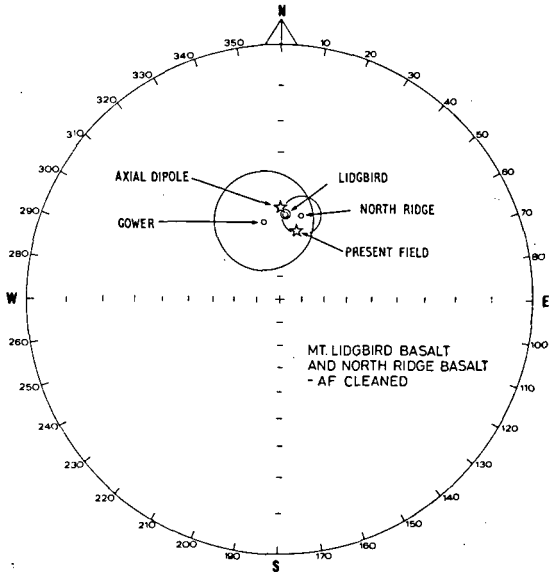


Fig. 13. Average directions of magnetisation and α_{95}s for the North Ridge Basalt lavas (17 flows), and flows from the Mount Lidgbird Basalt. Results from the Mount Lidgbird Basalt are divided into two groups, samples from the lower part of the sequence labelled Lidgbird, and samples from the ridge leading to the top of Mount Gower, affected by lightning, labelled Gower. All directions are normal (upward pointing).

Data from the first two areas will be discussed together, as they form a coherent group from a stratigraphic thickness of about 200 m, whereas the data from the ridge leading to Mount Gower will be considered separately, as they have different magnetic characteristics. Results from the former group show extremely small confidence circles for both within-flow (Table 5) and between-flow (Table 6) data. All have normal polarity and the mean direction is close to that of the axial dipole field (Fig. 13). The α_{95} is less than 2°, which on any published model of secular variation has to be interpreted as indicating a very short time for the extrusion of these lavas, perhaps only a few hundred years.

Data from the lavas collected on the ridge leading to Mount Gower show excessive scatter and high intensities of magnetisation, although all sites have normal polarity (Table 5). The high intensity of magnetisation was noted in the field, as the magnetic compass was commonly deflected dramatically as an outcrop was approached. The samples often were difficult to clean in AF fields. We conclude that virtually all these flows have been affected by lightning strikes; this is not unreasonable as all the outcrops are in a very exposed position on the ridge. The dyke sampled from this ridge also has normal polarity, but yields very scattered directions.

Discussion

The palaeomagnetic data on rocks from Lord Howe Island are of particular value in the interpretation of the geology of the island. The results show that the North Ridge Basalt and the Mount Lidgbird Basalt were erupted during periods when the earth's magnetic field had normal polarity, and that the two phases of eruption were separated by at least one interval of reversed polarity, as recorded by the dykes, sheets and the Boat Harbour Breccia. There are a number of intervals of normal and reversed polarity in the period 7.0 to 6.3 Ma ago (cf. La Brecque et al., 1977), allowing for the 2.66 percent increase in age of polarity boundaries required by the newly recommended ^{40}K decay constants (Steiger & Jäger, 1977) used in this paper, so that there is no difficulty in fitting the Lord Howe Island data into the polarity time scale. The age data, however, are not sufficiently precise to enable correlation of the lavas or intrusive rocks on Lord Howe Island with particular intervals of normal or reversed polarity of the geomagnetic polarity time scale.

The combined directions of magnetisation for the North Ridge Basalt, the dykes and sheets (correcting for polarity) and the Mount Lidgbird Basalt — 71 units in all—are distinguishable by 1° only, at the 95 percent confidence level, from the direction of the axial dipole field. We conclude from this that there is little evidence for latitudinal movement of Lord Howe Island relative to the spin axis of the earth, on the assumption that the geomagnetic field on average conforms to an axial dipole parallel to the spin axis.

The palaeomagnetic data are of considerable significance in providing confirmation that the dykes and sheets in the Boat Harbour Breccia have not suffered tectonic tilt, as had been proposed by Game (1970). The degree to which the Boat Harbour Breccia is magnetically overprinted lends support to the notion that this unit formed in close proximity to a substantial source of heat, presumably the throat of the volcano.

Finally the remarkably small dispersion of directions of magnetisation of samples from the Mount Lidgbird Basalt from sites unaffected by lightning strikes, suggests very rapid eruption of these basalt flows, consistent with the interpretation that the lavas are filling a caldera.

EVOLUTION

Lord Howe Island is the erosional remnant of a large shield volcano that was built above sealevel in the late Miocene. The island is but the subaerial part of a much larger structure, more than 25 km in diameter, which was built up from ocean floor more than 1800 m deep. Virtually nothing is known about the submerged pedestal of Lord Howe Island but we have little doubt that it is composed of products of submarine volcanism.

The earliest rocks recognized on Lord Howe Island are those of the Roach Island Tuff, formed by explosive activity, probably as a result of interactions of magma with seawater as the volcanism breached sealevel. In due course a major vent built itself above sealevel, probably in the general vicinity of the present Mount Lidgbird, and lavas erupted from this centre constructed a large shield volcano to at least an elevation of 900 m and possibly a good deal higher. This subaerial shield building phase of volcanism produced flows of tholeiitic affinities, comprising the North Ridge Basalt, about 6.9 Ma ago.

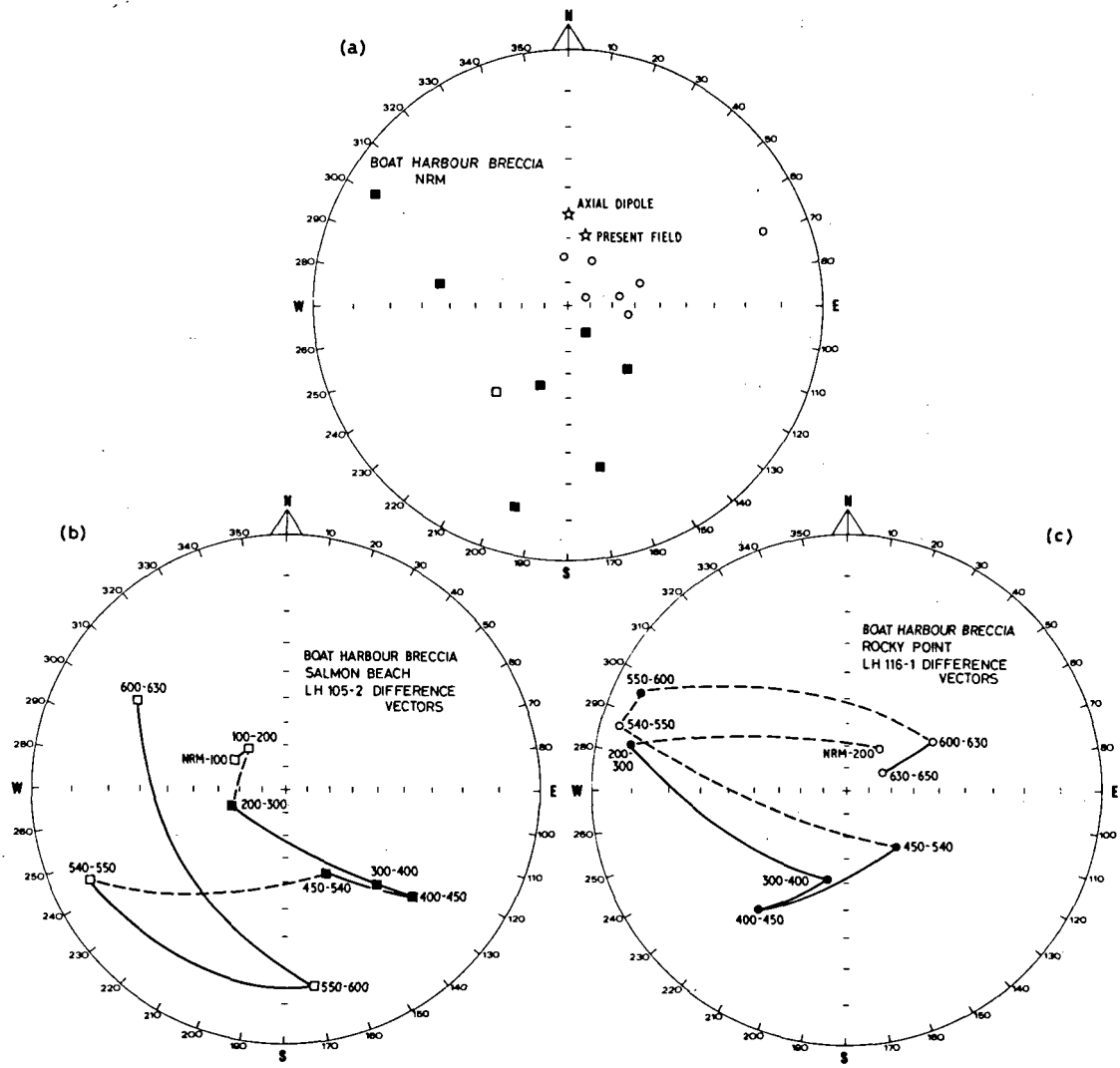


Fig. 14. Equal-angle stereographic plots showing directions of magnetisation in the Boat Harbour Breccia. A. NRM measured directions; circles represent specimens from Rocky Point area; squares represent specimens from Salmon Beach. B and C. Examples of difference vectors calculated in conjunction with thermal demagnetisation. Open (closed) symbols represent upward (downward) pointing magnetisation directions.

We suggest that the Boat Harbour Breccia, probably mainly agglomerate, may have formed subsequently by a major explosive episode centred on the main vent of the volcano. The emplacement of a great dyke and cone sheet swarm then took place, possibly because of dilation of the whole central region of the volcano above a large magma chamber, from which the magma was derived to form the dykes and sheets. Subsequently collapse occurred in the vicinity of the summit of the volcano, producing a caldera at least 900 m deep with dimensions exceeding 5 km by 2 km. This caldera was filled rapidly some 6.4 Ma ago by successive lava flows comprising the Mount Lidgbird Basalt of mildly alkali composition. This brought to a close the volcanic history of Lord Howe Island, after a period of activity in the order of 0.6 Ma during the late Miocene. Subsequently erosional forces have removed much of the shield volcano, leaving only remnants of the northern flanks. The southern, western and eastern slopes of the constructional dome have been completely eroded away to expose the caldera filling lava sequence. During periods of lower sealevel in the Pleistocene and possibly late Pliocene a large bevelled platform was exposed around the island and served as a source for calcareous material that was carried by wind and deposited around the eroded volcanic hills of Lord Howe Island as the Neds Beach Calcarenite.

Since formation of the calcarenite the island has continued to be modified by erosion, although the

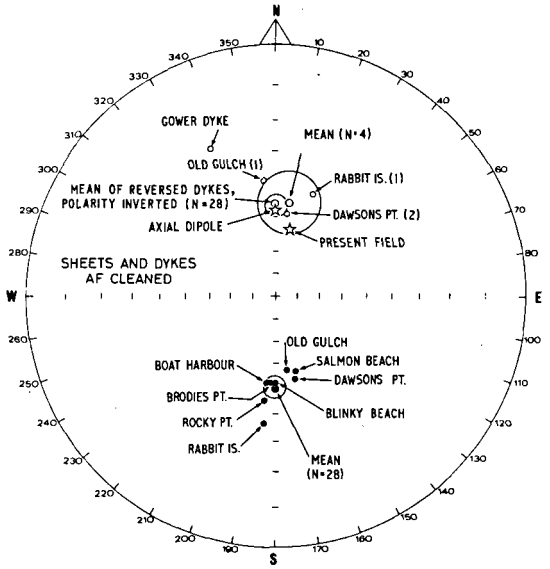


Fig. 15. Direction of magnetisation of Lord Howe Island dykes and sheets averaged by locality. a_{95} for each mean is represented by a circle on the equal-angle projection.

changes are small in comparison with those that occurred during the long interval between cessation of volcanism and deposition of the calcarenite.

The history of Lord Howe Island volcano is similar to many other volcanoes formed in the oceanic environment. In particular, the volcano had a relatively short period of activity, substantially less than 1 Ma, a feature common to many oceanic volcanoes (cf. McDougall, 1964; Duncan & McDougall, 1976; McDougall & Duncan, 1980).

LORD HOWE SEAMOUNT CHAIN

A line of reefs, guyots and banks extends about 1000 km due north from Lord Howe Island (Fig. 1). These steepsided seamounts surely are of volcanic origin, as has been suggested by earlier workers (Standard, 1961; Conolly, 1969; Vogt & Conolly, 1971; Slater & Goodwin, 1973). The summits of the seamounts are broad platforms capped by coral reefs or limestone (Slater & Goodwin, 1973). Elizabeth and Middleton Reefs are at sealevel, whereas Gifford Guyot, Capel Bank and Kelso Bank are submarine, shoaling to within 290 m, 55 m and 18 m of sealevel respectively (Slater & Goodwin, 1973).

The Tasmantid Guyots also are aligned north-south and occur about 300 km to the W of the Lord Howe seamount chain; they are built upon oceanic crust in the Tasman Basin from depths of more than 4500 m (Standard, 1961; Conolly, 1969; Vogt & Conolly, 1971). Limestone is found on the flat summits of the guyots but dredging on Brittania, Derwent Hunter and Barcoo Guyots and on Taupo Bank also reveals the presence of basalt, confirming their volcanic origin (David, 1932; Slater & Goodwin, 1973).

Volcanic island and seamount chains are relatively common in the ocean basins. Each chain is thought to record the movement of the lithospheric plate over a magma source located below the plate in the upper mantle (Wilson, 1963), or perhaps even deeper within the earth (Morgan, 1971; 1972). Volcanism occurs above the hot spot or plume, and as the volcano so produced is moved laterally away by plate motion it becomes disconnected from its magma source. A new volcano may develop in place of the one carried away on the lithospheric plate. Thus a line of volcanoes is produced which shows a monotonic progression of age along it. Vogt & Conolly (1971) proposed an origin of this kind for the Tasmantid Guyots and the Lord Howe seamount chain, and from the observation that the summits of the Tasmantid Guyots are progressively further below sealevel northward they inferred an increase in age of the guyots toward the north. Thus they concluded that the Australian lithospheric plate, on which the guyots are built, has moved in a northerly direction over the underlying mantle in which the magma sources for the two seamount chains are located. Under this hypothesis the seamount chains should trace out small circles about the common pole of rotation for the Australian and Antarctica plates. As the seamount chains are nearly straight, rather than curved, this suggests that they must be nearly equatorial relative to the rotation pole. From marine magnetic anomaly and other data Weissel et al. (1977) derived the pole of rotation for the Australian and Antarctica plates for the interval early to middle Miocene at about 11° north latitude, 35° east longitude, so that the seamount chains in the Tasman Sea are within 35° of an equatorial location relative to this pole of rotation, reasonably consistent with their linear, rather than curved trends.

Although palaeomagnetic data from igneous rocks in eastern Australian record a general northward migration of Australia over the last 100 Ma, little northward component of movement is indicated for the last 20 Ma (Wellman et al., 1969). However, studies of marine magnetic anomalies in the Southern Ocean show that Australia began to separate from Antarctica about 55 Ma ago and this has continued to the present (Weissel & Hayes, 1972; Weissel et al., 1977). These authors derive an average rate of separation of the two continents of about 5.6 cm/a. That this motion is taken up by northward movement of the Australian plate from Antarctica, with the latter essentially stationary, is supported by the occurrence in eastern Australia of a broad band of central volcanoes that become younger to the south (Fig. 1). These volcanoes range in age from 33 Ma to 6 Ma, and the rate of migration of the centre of volcanism is estimated to be 6.6 (±0.5) cm/a to the south, on the assumption that the magma source below the Australian plate is fixed (Wellman & McDougall, 1974). This translates into northward movement of 6.6 cm/a for the Australian plate, very similar to the average rate of separation of Australia and Antarctica derived from the marine magnetic anomaly data. Applying a rate of northward movement of the Australian plate of 6 cm/a, arrived at by averaging the two estimates just given, it is possible to predict the age of the seamounts of the Lord Howe chain, as the average age of Lord Howe Island itself is 6.7 Ma. On this basis the following approximate ages are predicted; Elizabeth Reef, 10.2 Ma; Middleton Reef,

10.7 Ma; Gifford Guyot, 15.6 Ma; Capel Bank, 18.4 Ma; Kelso Bank, 20.4 Ma; Argo Bank, 21.8 Ma and Nova Bank 23 Ma. Similarly, if volcanism were to resume now in the Lord Howe seamount chain, we predict that its location would be approximately 400 km S of Lord Howe Island. Whether Flinders Seamount, about 350 km nearly due S (S 10° E) of Lord Howe Island (Fig. 1), is the site of relatively youthful volcanism cannot be ascertained from presently available data. This seamount rises at least 700 m above surrounding seafloor, reaching to within 1 750 m of sealevel (van der Linden, 1968).

Predictions as to the age of the seamounts N of Lord Howe Island could be tested if dredging recovered volcanic samples suitable for age determination. However, because of coral cappings on these seamounts it would be difficult to obtain volcanic material, as shown already by Slater & Goodwin (1973). As demonstrated by these authors, a more practical proposition would be further dredging of the Tasmantid Guyots for volcanic samples, in order to determine the ages of these seamounts and to determine the rate of migration of the volcanism. Such data would be valuable in providing constraints on the origin of the seamount chain and the evolution of the Tasman Basin.

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Note added in proof

Since this paper was submitted for publication, two papers that contain relevant information have come to the attention of the authors. Ploshko et al. (1979) presented data on the mineralogy of dykes intruding the North Ridge Basalt and on the Mount Lidgbird Basalt, and Willcox et al. (1980) reported geophysical results from and adjacent to the Lord Howe Rise.

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